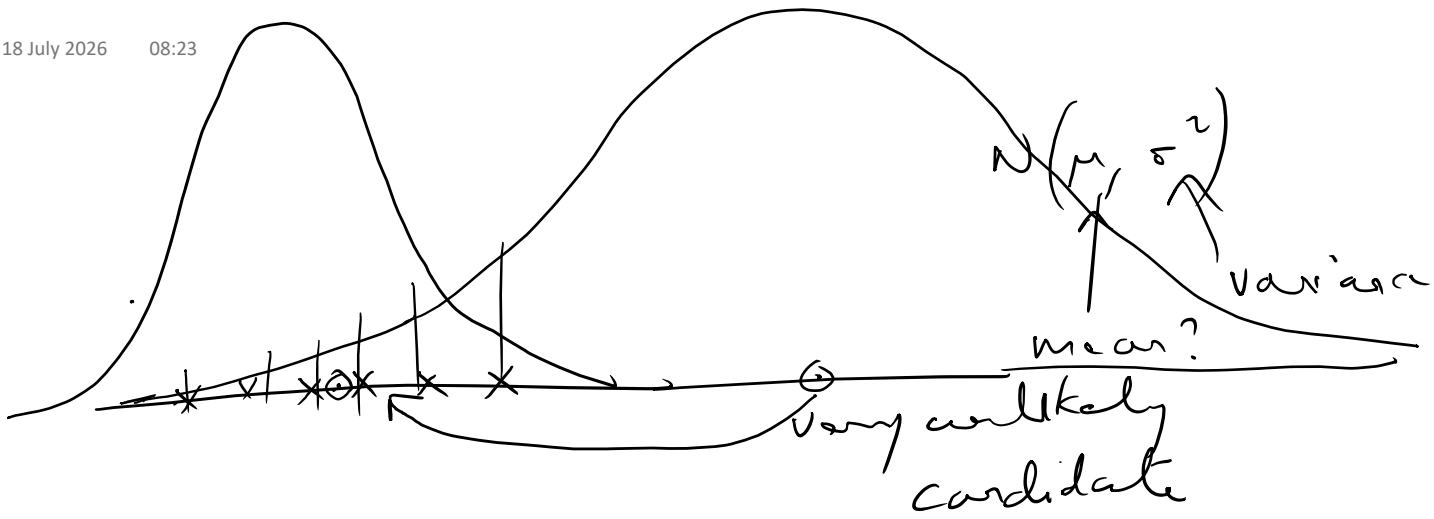


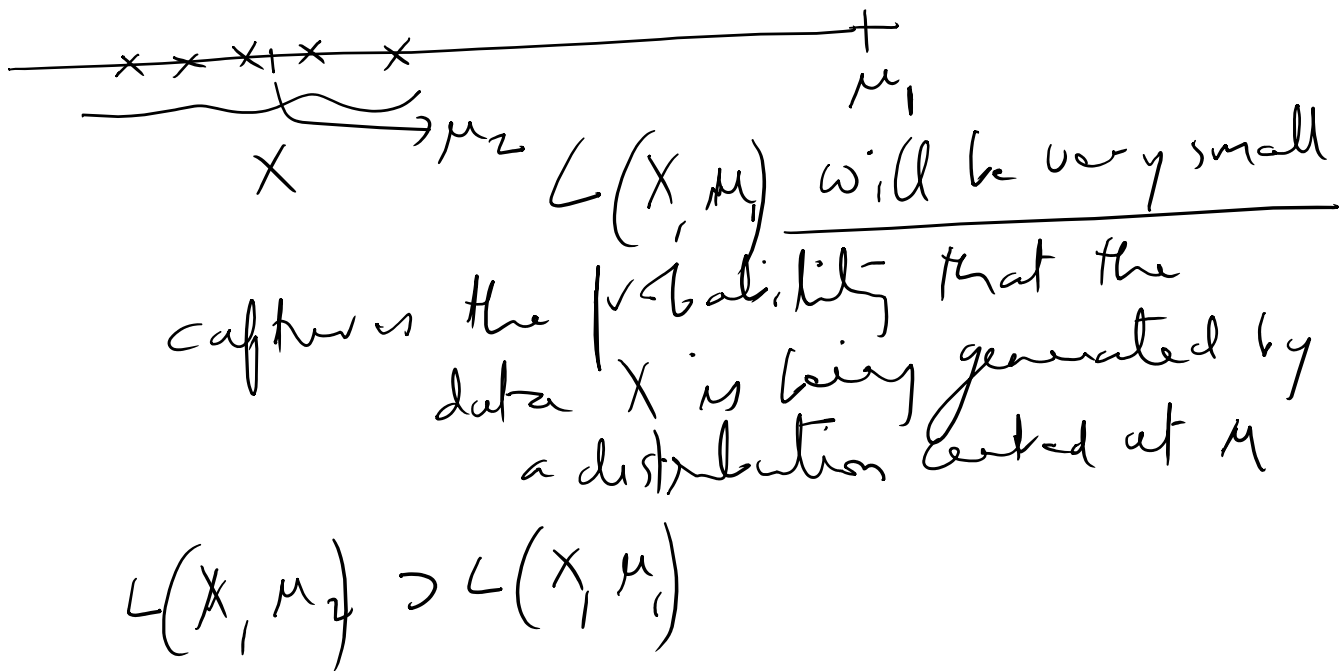
— — — — — observed
1 2 3 . . . sample

what is the most likely value of
the probability parameter that
led to this particular selection



maximize the probability of
generating the given samples

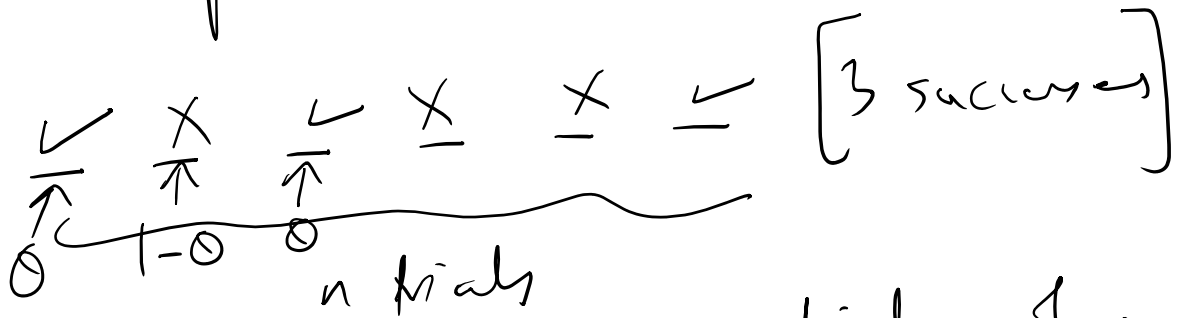
$L(\theta)$ = a function in terms of the parameter θ .



We study $\log L(\theta)$ rather than $L(\theta)$
because $\log$ is a monotonic function

Let $\log L(\theta)$ be maximized at $\theta = \theta_{\max}$
Then $L(\theta)$ is also maximized at $\theta = \theta_{\max}$

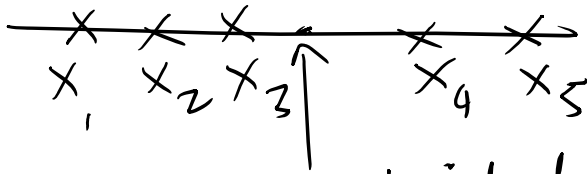
What does $\binom{n}{k} \theta^k (1-\theta)^{n-k}$ actually capture



particular string of n trials where there were exactly 3 successes as marked in specific positions
 (Generating this particular string) = $\theta^3 (1-\theta)^{n-3}$

$\binom{n}{3} \theta^3 (1-\theta)^{n-3} \rightarrow$ is the total probability of generating all possible strings like the one above

$$\binom{n}{3} \theta^3 (1-\theta)^{n-3} \rightarrow \begin{cases} \underline{\checkmark} \quad \underline{\times} \quad \underline{\checkmark} \quad \underline{\times} \quad \underline{\checkmark} \\ \underline{\checkmark} \quad \underline{\checkmark} \quad \underline{\times} \quad \underline{\times} \quad \underline{\times} \\ \underline{\times} \quad \underline{\times} \quad \underline{\times} \quad \underline{\checkmark} \quad \underline{\checkmark} \\ \vdots \end{cases}$$



most likely $\mu = \frac{x_1 + x_2 + \dots + x_5}{5}$

most likely $\hat{\sigma}^2$ is not the sample variance

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

biased
estimate of
sample
variance

Actual estimate of sample variance = $\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

$$F = \frac{S_1^2}{S_2^2}$$

S_1 is a sample ^{variance} from a normal distribution with mean μ , and std dev σ , consisting of n_1 instances

S_2 has n_2 instances.

$$S_1 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (x_i - \bar{x})^2 \rightarrow \text{from } S_1$$

$$S_2 = \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (x_i - \bar{x})^2 \rightarrow \text{from } S_2$$

ANOVA (analysis of variance) is a family of statistical tests used to determine whether the means of several populations differ significantly

$$\begin{array}{|l} H_0: \mu_1 = \mu_2 = \dots = \mu_K \\ H_1: \text{at least two of these means differ} \end{array}$$

Why do we need ANOVA?

Why not use multiple t-tests?

Suppose you have 3 groups - Group A, Group B, Group C.

Suppose you want to determine whether their means differ

t-tests: A vs B, A vs C, B vs C
requires 3 tests.

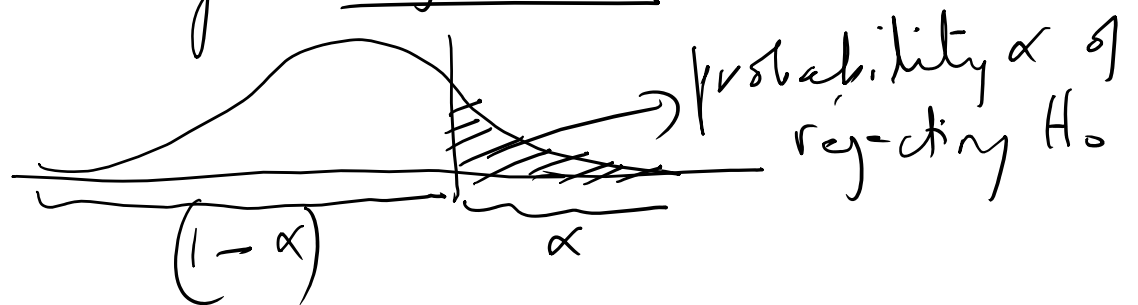
for 4 groups, we will need $\binom{4}{2} = 6$ tests

for 10 groups, we will need $\binom{10}{2} = 45$ tests

The problem is that every t-test has a probability of making a type I error and these errors compound

What is a type I error?

↳ when the null hypothesis is true and you reject H_0



Where does the error confounding come from

if each test uses α (say equal to 0.05)
the probability of at least one
false positive (type I error) increases
rapidly

For m independent tests

$P(\text{at least one false positive})$

$= 1 - \text{Zero false positives}$

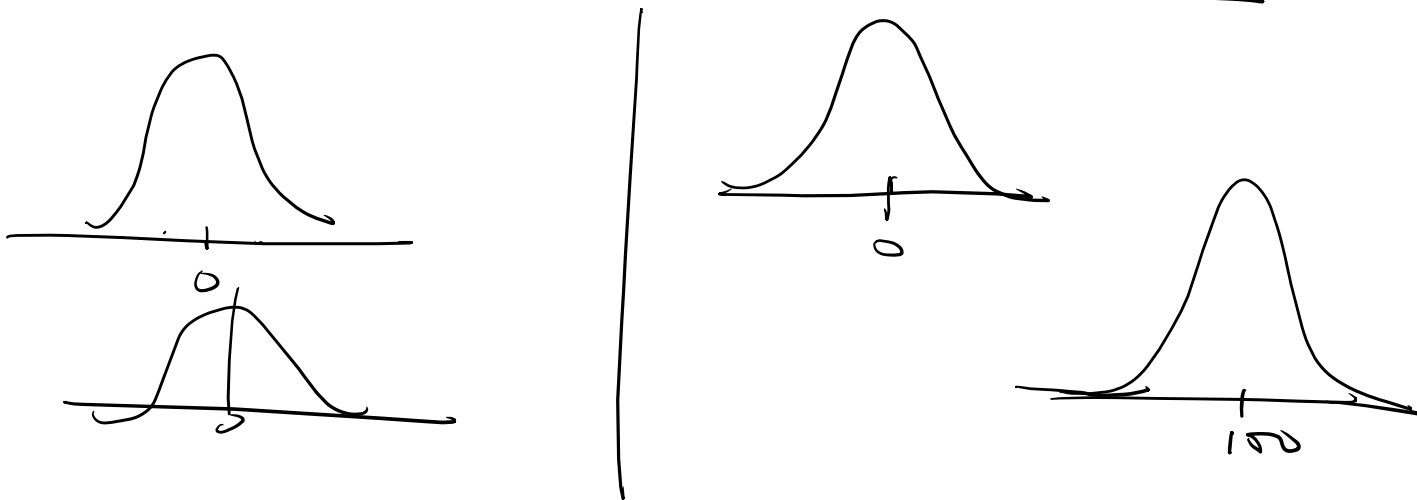
$$= 1 - (1 - \alpha)^m$$

for 10 pairwise tests, $1 - (1 - 0.05)^{10}$
 $= 1 - 0.95^{10}$

$$\approx \underline{\underline{0.40}}$$

$P(\text{at least one type I error}) \approx 0.40$

The key idea of ANOVA is that if population means differ, the variation between groups should be much larger than the variation within groups



Unlike multiple t-tests we perform one overall hypothesis test

One-way ANOVA studies the effect of
one categorical factor on a numerical
response

Example: Does fertilizer type affect
crop yield?

factor: fertilizer

Levels: A, B, C, D

Response: Crop yield

Question is: are all fertilizer means
equal?

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k$$

H_1 : at least one mean differs

[this does not tell us which mean differs]

Suppose every group has the same mean.
Then most of the variation should come from random noise within each group

if one group's mean is much larger than another's then the variation between group means becomes large

Suppose there are K groups and N observations in total

Let $\bar{y}$ be the overall mean

Let $\bar{y}_i$ be the mean of group i

The total variation is

$$SST = \sum_{i=1}^K \sum_{j=1}^{n_i} (y_{ij} - \bar{y})^2$$

ANOVA writes $SST = SSB + SSW$

$$SSB = \sum_{i=1}^K n_i (\bar{y}_i - \bar{y})^2 \left[\begin{array}{l} \text{Between group} \\ \text{sum-of-squares} \end{array} \right]$$

$$SSW = \sum_{i=1}^K \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2 \left[\begin{array}{l} \text{within group} \\ \text{sum of squares} \end{array} \right]$$

$[n_i = \# \text{ of instances in group } i]$

$$SST = \sum_{i=1}^K \sum_{j=1}^{n_i} (y_{ij} - \bar{y})^2$$

$$y_{ij} - \bar{y} = (y_{ij} - \bar{y}_i) + (\bar{y}_i - \bar{y})$$

$$\begin{aligned} (y_{ij} - \bar{y})^2 &= \left[(y_{ij} - \bar{y}_i) + (\bar{y}_i - \bar{y}) \right]^2 \\ &= (y_{ij} - \bar{y}_i)^2 + 2(y_{ij} - \bar{y}_i)(\bar{y}_i - \bar{y}) \\ &\quad + (\bar{y}_i - \bar{y})^2 \end{aligned}$$

$$\begin{aligned} SST = \sum_i \sum_j (y_{ij} - \bar{y})^2 &= \underbrace{\sum_i \sum_j (y_{ij} - \bar{y}_i)^2}_{SSW} \\ &\quad + \underbrace{\sum_i \sum_j 2(y_{ij} - \bar{y}_i)(\bar{y}_i - \bar{y})}_{0} \\ &\quad + \underbrace{\sum_i \sum_j (\bar{y}_i - \bar{y})^2}_{SSB} \end{aligned}$$

$$\sum_{i=1}^K \sum_{j=1}^{n_i} (\bar{y}_i - \bar{y})^2 = n_i (\bar{y}_i - \bar{y})^2 = \underline{SSB}$$

$$\sum_i \sum_j 2(y_{ij} - \bar{y}_i)(\bar{y}_i - \bar{y})$$

$$\sum_i \sum_j 2(y_{ij} - \bar{y}_i)(\bar{y}_i - \bar{y})$$

$$= \sum_{i=1}^K (\bar{y}_i - \bar{y}) \left(\sum_{j=1}^{n_i} 2(y_{ij} - \bar{y}_i) \right)$$

$$\sum_{j=1}^{n_i} y_{ij} - \bar{y}_i = 0$$

$$\sum_{j=1}^{n_i} y_{ij} = \underline{n_i \bar{y}_i}$$

$$\sum_{j=1}^{n_i} \bar{y}_i = \underline{n_i \bar{y}_i}$$

Between groups $\underline{MSB} = \frac{\textcircled{SSB}}{(K-1)}$

Within groups $MSW = \frac{SSW}{N-K}$

ANOVA test statistic

$$F = \frac{MSB}{MSW}$$

if the null hypothesis is true
 $MSB \approx MSW$ and $F \approx 1$

if the group means differ $F > 1$

$$SSB = \sum_{i=1}^K n_i (\bar{y}_i - \bar{\bar{y}})^2$$

There are K group means $\bar{y}_i$ but
they are not independent

$$\bar{\bar{y}} = \frac{1}{N} \sum_i n_i \bar{y}_i$$

Therefore only $(K-1)$ group mean
deviations vary freely

$$\underline{df(SSB) = K - 1.}$$

We can also show $df(SSW) = N - K$

$$SSW = \sum_{i=1}^K \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$$

Each $\bar{y}_i$ satisfies

$$\sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i) = 0$$

There are only $(n_i - 1)$ deviations that are free

$$\sum_{i=1}^K (n_i - 1) = \sum_{i=1}^K n_i - K = \underline{N - K}$$

Two-way ANOVA is a natural extension of one-way ANOVA.

It asks whether two factors affect the response and whether they interact

	Drip	Sprinkler
Fertilizer (A1)	several plots	several plots
Fertilizer (A2)	several plots	several plots
Fertilizer (A3)		

The factors are fertilizer and irrigation
 Each cell contains several observations

We are interested in 3 hypotheses:

(a) Main effect of fertilizer
 Do fertilizers have different
 mean yields?

$$H_0^{(A)} : \mu_{1.} = \mu_{2.} = \mu_{3.}$$

(b) Main effect of irrigation

$$H_0^{(B)} : \mu_{.1} = \mu_{.2}$$

(c) interaction

Does the effect of fertilizer depend on irrigation

Example of no interaction

	Drip	Sprinkler
A1	20	15
A2	30	25
A3	40	35

Moving from sprinkler to drip always adds 5. $\rightarrow$ no interaction

Example of interaction

	Drip	Sprinkle
A1	20	18
A2	35	22
A3	55	25

A3 performs
dramatically
better
under drip
irrigation

Statistical model

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

α_i = effect of fertilizer

β_j = effect of irrigation

$(\alpha\beta)_{ij}$ = interaction effect

ϵ_{ijk} = random error

$$SST = \sum (y_{ijk} - \bar{y})^2$$

(total variation)

$$SST = SSA + SSB + SSAB + SSE$$

SSA = variation explained by factor A

SSB = variation explained by factor B

$SSAB$ → interaction

SSE → within cell variation

$$SST = SSA + SSB + SSAB + SSE$$

$$\bar{y}_{ij} = \text{mean of cell } (i, j)$$

$$\bar{y}_{i.} = \text{Mean of row } i$$

$$\bar{y}_{.j} = \text{mean of column } j$$

$$\begin{aligned} \bar{y}_{..} &= \text{grand mean} = \\ &= \frac{1}{abn} \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n y_{ijk} \end{aligned}$$

$$\begin{aligned} \text{factor A} \quad SSA &= bn \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 \\ \hline SSB &= an \sum_{j=1}^b (\bar{y}_{.j} - \bar{y}_{..})^2 \end{aligned}$$

$$SS_{AB} = n \sum_i \sum_j (\bar{y}_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..})^2$$

$$SSE = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij})^2$$

$$MS_A = \frac{SS_A}{a-1}$$

$$MS_B = \frac{SS_B}{b-1}$$

$$MS_{AB} = \frac{SS_{AB}}{(a-1)(b-1)}$$

$$MS_E = \frac{SSE}{ab(a-1)}$$

F - tests

$$F_A = \frac{MS_A}{MS_E}$$

$$F_B = \frac{MS_B}{MS_E}$$

$$F_C = \frac{MS_{AB}}{MS_E}$$

Heads up about covariance

$X, Y \rightarrow$ two random variables
how do they vary together?

Is there a relationship?

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

if X and Y are independent

$$E(XY) = E(X)E(Y)$$

$$\underline{\text{Cov}(X, Y) = 0}$$

$$\text{Cov}(X, X) = E(X^2) - E^2(X) = \underline{\text{Var}(X)}$$

how can we estimate covariance
from a sample

Suppose we have paired observations
of the two variables

$(x_1, y_1) \quad (x_2, y_2) \quad \dots \quad (x_n, y_n)$

$$\bar{x} = \frac{\sum x_i}{n}$$

$$\bar{y} = \frac{\sum y_i}{n}$$

Std estimator for covariance

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

Sample covariance

$$E[S_{XY}] = \text{Cov}(X, Y) = \sigma_{XY}$$

Sample covariance is the dot product
of two vectors

$$\begin{bmatrix} x_1 - \bar{x} \\ x_2 - \bar{x} \\ \vdots \\ x_n - \bar{x} \end{bmatrix}_{n \times 1}$$

$$\begin{bmatrix} y_1 - \bar{y} \\ y_2 - \bar{y} \\ \vdots \\ y_n - \bar{y} \end{bmatrix}_{n \times 1}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

Covariance has units

Correlation coefficient

$$r = \frac{S_{xy}}{S_x S_y}$$

$$S_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$S_x = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

$$S_y = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2}$$

r is like cosine of an angle between two vectors.

$$\underline{-1 \leq \cos \theta \leq 1}$$

$$\underline{-1 \leq r \leq 1} \rightarrow \underline{\text{Correlation coefficient}}$$

Vector 1: $\frac{1}{\sqrt{n-1}} \begin{bmatrix} x_1 - \bar{x} \\ x_2 - \bar{x} \\ \vdots \\ x_n - \bar{x} \end{bmatrix}$

Vector 2: $\frac{1}{\sqrt{n-1}} \begin{bmatrix} y_1 - \bar{y} \\ y_2 - \bar{y} \\ \vdots \\ y_n - \bar{y} \end{bmatrix}$

$$S_{xy} =$$

dot product
of vector 1 and
vector 2

$$S_x = \text{length of vector 1}, \quad S_y = \text{length of vector 2}$$